

ATHARVA R ANKAD

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PROFESSIONAL SUMMARY

Third year Integrated M.Tech CSE student specializing in Computational & Data Science with hands-on experience building **end-to-end ML pipelines and scalable retrieval systems (RAG)**. Recognized for developing high-impact, data-driven prediction models with a core focus on NLP and backend system design.

EDUCATION

VIT Bhopal University	Bhopal, India
<i>Integrated M.Tech Computer Science and Engineering (Computational and Data Science) – CGPA: 7.83</i>	<i>2023-2028</i>
Sri Chaitanya Techno School	Bangalore, India
<i>CBSE Class XII Science – 75.6%</i>	<i>2021-2023</i>
HAL Gnanajyoti School	Bangalore, India
<i>ICSE Class X – 89.8%</i>	<i>2009-2021</i>

TECHNICAL SKILLS

Languages & Core: Python, Java, Git, Linux
Machine Learning & NLP: scikit-learn, LangChain, Sentence Transformers, Vector Search, LLM Integration
Data Processing & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Hadoop (Fundamentals)
Backend & Databases: FastAPI, REST API Architecture, MongoDB, MySQL

PROJECTS

- Fasal Sahayak AI** | *Python, FastAPI, scikit-learn, RandomForest, Pandas, NASA POWER API* [GitHub](#)
- Built an end-to-end ML pipeline and REST API for **data-driven crop yield prediction and agronomic advisory, integrating 13 heterogeneous data sources** including 16K+ historical crop records, live meteorological data (NASA POWER API), and soil research from ICAR and government publications.
 - Designed a **two-stage RandomForestRegressor pipeline** separating outlier crops (e.g., sugarcane) from annual crops, applying log1p target normalization to handle skewed yield distributions.
 - Developed a **dual-mode advisory engine** providing generalized agronomic guidance for standard users and threshold-based warnings for farmers entering Soil Health Card (SHC) metrics.
 - Reduced query latency by **~40%** by implementing **lightweight JSON lookup stores** for pre-computed geographic data, eliminating runtime database calls.
- RAG Semantic Search System** | *Python, LangChain, SentenceTransformers, MongoDB, Groq API* [GitHub](#)
- Built a **Retrieval-Augmented Generation (RAG)** pipeline that ingests PDF documents, generates local semantic embeddings using **SentenceTransformers (all-MiniLM-L6-v2)**, and performs vector similarity search via **MongoDB Atlas Vector Search**.
 - Implemented context-aware **smart chunking** with LangChain text splitters to preserve paragraph structure and semantic meaning across document chunks.
 - Integrated **Groq API** with a strict contextual system prompt to ensure LLM responses are grounded solely in retrieved document context, with citation support including page numbers and relevance scores.

ACHIEVEMENTS

Smart India Hackathon (SIH) 2025 — Internal Nominee	VIT Bhopal University
<i>Selected among the top 45 out of 949 submissions in the university's internal round.</i>	<i>2025</i>

CERTIFICATIONS

Google IT Support Certificate	Google
<i>Foundations of IT support, networking, and systems administration</i>	<i>2026</i>
Building RAG Apps using MongoDB	MongoDB
<i>Credential in Retrieval-Augmented Generation application development</i>	<i>2026</i>